

[3] Retrieval-Augmented Generation for Educational Application: A Systematic Survey

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1-Minute Executive Takeaway

Key Finding: Highlights that computational cost and dynamic knowledge updating are the primary barriers to deploying educational RAG.

Core Research Objective

To systematically survey the emerging landscape of retrieval-augmented generation applications across educational contexts, identifying pedagogical opportunities, technical architectures, and open challenges.

Methodology & Architecture

Systematic literature review covering interactive learning companions, automated assessment systems, content curation pipelines, and institutional tutoring platforms across K-12 and higher education.

Key Empirical Findings & Evidence

Identified knowledge-base updating latency, computational resource constraints in educational institutions, and curricular hallucinations as the three foremost technical bottlenecks impeding widespread RAG adoption in schools and universities.

Direct Relevance to Our Kwiz Paper

Key motivating literature in Section 1, Section 2.1, and Section 2.5. Our Kwiz paper directly resolves the exact challenges identified by Li et al. by proposing incremental SHA-256 chunk caching and a single-GPU self-hosted architecture for Moodle.